

Interpretable Multi-Source Machine Learning for Urban Classification in Data-Scarce Regions

Peter Edemewe Ugege¹, John Aigbokhan², Emmanuel Akalumhe³

¹Informatics and ICT Resource Centre, Forestry Research Institute of Nigeria, Ibadan

²Environmental Modeling and Biometrics Department, Forestry Research Institute of Nigeria, Ibadan.

³Informatics and ICT Resource Centre, Forestry Research Institute of Nigeria, Ibadan

Abstract

Rapid urban expansion in data-scarce regions demands accurate and interpretable methods for monitoring urban growth. This study presents an interpretable multi-source machine learning framework for urban classification in Lagos, Nigeria, integrating spectral, socio-economic, and temporal geospatial data. A dataset of 36,000 samples spanning 2000–2024 was constructed using spectral bands, derived indices, night-time lights, population density, and elevation. We evaluate Random Forest, XGBoost, and a multilayer perceptron, achieving consistent performance across models (accuracy = 0.87, AUC > 0.94). Notably, results show that feature integration plays a more critical role than model complexity, with ensemble methods performing comparably to neural networks. To enhance interpretability, SHAP-based explainable AI is applied to quantify feature contributions. The analysis identifies night-time lights and population density as dominant drivers of urban classification, while spectral variables provide complementary discriminatory power. These findings demonstrate that combining heterogeneous data sources with interpretable machine learning enables both accurate prediction and meaningful insight. Overall, the proposed framework provides a scalable and transparent approach for urban monitoring in rapidly developing, data-limited environments.

1 Introduction

Urbanization is one of the most significant forms of land transformation occurring globally, with profound implications for environmental sustainability, infrastructure planning, and socio-economic development. Rapid urban growth in developing regions, particularly in sub-Saharan Africa, has intensified the need for accurate, scalable, and interpretable methods for monitoring urban expansion [Seto *et al.*, 2012; United Nations Human Settlements Programme, 2022]. Lagos, Nigeria, represents one of the fastest growing megacities in Africa, experiencing substantial population growth and spatial expansion over recent decades [United Nations Human Settlements Programme, 2022]. Understanding these

dynamics requires robust approaches capable of integrating diverse data sources and capturing complex urban patterns.

Remote sensing has become an indispensable tool for mapping urban land cover due to its ability to provide consistent spatial observations across large geographic areas. Traditional classification approaches based on spectral thresholds or statistical classifiers have been widely applied [Yu and Fang, 2023; Weng, 2012]. However, urban environments are inherently heterogeneous, often consisting of mixtures of built-up surfaces, vegetation, and bare soil, which limits the effectiveness of conventional methods.

Machine learning methods have significantly improved classification performance by modeling nonlinear relationships in geospatial data. Ensemble approaches such as Random Forest are widely used due to their robustness and ability to handle high-dimensional features [Belgiu and Drăguț, 2016]. Gradient boosting methods, particularly XGBoost, further enhance predictive performance through sequential optimization [Chen and Guestrin, 2016]. In parallel, deep learning models have demonstrated strong capability in capturing complex feature interactions and extracting patterns from large datasets [L. Ma and Johnson, 2019].

Recent studies have emphasized the integration of multi-source datasets, including spectral indices, night-time lights, and population data, to improve urban detection. Night-time lights have been shown to correlate strongly with urbanization and economic activity, while population density provides complementary information on human settlement patterns [Li and Zhou, 2016; Tatem, 2017]. These heterogeneous data sources enable more comprehensive modeling of urban dynamics

Despite advances in predictive modeling, interpretability remains a key challenge in applying machine learning to urban studies. Explainable Artificial Intelligence (XAI) techniques provide a principled approach for understanding model predictions. Among these, SHapley Additive exPlanations (SHAP) offers a unified framework for quantifying feature contributions based on cooperative game theory [Lundberg and Lee., 2017]. SHAP enables both global and local interpretability, making it particularly suitable for geospatial applications where understanding the influence of environmental and socio-economic variables is critical.

However, relatively few studies have combined multi-source geospatial data, multiple machine learning paradigms,

and explainable AI within a unified framework for urban classification, particularly in rapidly growing African cities. Addressing this gap is essential for improving both predictive accuracy and interpretability in urban monitoring.

To address these challenges, this study proposes an interpretable machine learning framework for urban classification in Lagos, Nigeria using multisource geospatial data. The approach integrates ensemble learning methods, deep learning, and SHAP-based explainable AI to achieve both high predictive performance and meaningful interpretation of model outputs.

While prior studies have explored machine learning for urban classification, this work advances the field by explicitly linking predictive performance with interpretable insights in a unified multi-source framework.

Contributions. This paper makes the following contributions:

- We propose an interpretable multi-source learning framework that integrates heterogeneous geospatial, socio-economic, and temporal data for urban classification in data-scarce environments.
- We show that, in multisource geospatial settings, feature representation and data integration are more critical than model complexity, with ensemble methods performing comparably to neural networks.
- We provide a comparative analysis of Random Forest, XGBoost, and neural networks on tabular geospatial data, offering empirical evidence of their effectiveness for urban classification.
- We apply SHAP-based explainable AI to identify the key drivers of urbanization, revealing consistent relationships between night-time lights, population density, spectral features, and urban patterns.
- We demonstrate that combining predictive modeling with explainability enables a transparent and scalable decision-support framework for urban monitoring in data-limited regions.

2 Materials and Methods

2.1 Dataset Preparation

A multisource geospatial dataset was developed to support urban classification in Lagos, Nigeria. Samples were extracted for six time periods spanning 2000, 2005, 2010, 2015, 2020, and 2024, resulting in a total of 36,000 observations. The dataset was balanced to ensure equal representation of urban and non-urban classes, thereby minimizing classification bias. Urban labels were derived from the Global Human Settlement Layer (GHSL) built-up surface dataset, which provides globally consistent information on built-up areas.

Predictor variables were derived from a combination of satellite imagery, socio-economic datasets, and topographic information. These variables included spectral bands such as Blue, Green, Red, Near Infrared (NIR), and Shortwave Infrared (SWIR), as well as derived indices including the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built-up Index (NDBI). In addition,

night-time lights intensity and population density were incorporated to capture human activity patterns associated with urban development, while elevation was included to account for topographic influences. A temporal variable (Year) was also included to represent changes across time.

Spectral indices such as NDVI and NDBI were used to differentiate vegetation and built-up areas, consistent with established approaches in remote sensing-based urban mapping [Weng, 2012; Y. Zha and Ni, 2003]. Night-time lights and population datasets provide complementary information on anthropogenic activity and have been widely used as proxies for urbanization and economic activity [Tatem, 2017].

Prior to model training, the dataset was randomly split into training and testing subsets using an 80:20 ratio. This ensured that model performance could be evaluated on unseen data while maintaining sufficient samples for training. All analyses were implemented in Python using standard machine learning libraries in a Google Colab environment to ensure reproducibility.

2.2 Machine Learning Models

Three classification algorithms were implemented using Python in a Google Colab environment to evaluate their performance in urban classification. The first model, Random Forest, is an ensemble learning method that constructs multiple decision trees and aggregates their predictions to improve classification accuracy and reduce overfitting [Z. Sun and Liang., 2024]. The second model, Extreme Gradient Boosting (XGBoost), is a gradient boosting algorithm that sequentially builds decision trees to minimize prediction error and has demonstrated strong predictive performance in a wide range of geospatial applications [Chen and Guestrin, 2016]. The third model is a multilayer perceptron (MLP) neural network implemented in PyTorch, which represents the deep learning approach and is capable of capturing complex nonlinear relationships between predictor variables and urban classification [T. Man and Ignatov, 2024].

2.3 Model Evaluation

Model performance was assessed using classification accuracy and the Area Under the Receiver Operating Characteristic Curve (AUC). Accuracy measures the proportion of correctly classified samples, providing an overall indication of model performance, while AUC evaluates the model's ability to discriminate between urban and non-urban classes across different classification thresholds [Fawcett, 2005]. These metrics were used to compare the performance of the three models and determine their effectiveness in urban detection.

2.4 Explainable AI Analysis

To enhance model interpretability, SHapley Additive exPlanations (SHAP) were applied to quantify the contribution of each predictor variable to model predictions. SHAP is based on cooperative game theory and provides consistent and locally accurate feature attributions for machine learning models [Lundberg and Lee., 2017]. This approach enables both global and local interpretation of model behavior, allowing for a detailed understanding of the factors influencing classification outcomes. Several SHAP-based visualizations were

generated to support interpretation, including feature importance bar plots, beeswarm summary plots, and dependence plots. These visualizations provide insights into both the relative importance of predictor variables and the direction of their influence on urban classification, thereby helping to identify the key drivers of urban development [A. V. Ponce-Bobadilla and Stodtmann, 2024].

3 Results

3.1 Model Performance

The three classification models demonstrated similar predictive performance in distinguishing urban and non-urban areas (Table 1). All models achieved an overall classification accuracy of 0.87, indicating strong predictive capability for urban classification.

Model	Accuracy	AUC
Random Forest	0.87	0.942
XGBoost	0.87	0.943
Neural Network	0.87	0.940

Table 1: Comparison of classification models

The Receiver Operating Characteristic (ROC) curves further illustrate the performance of the models (Figure 1). XGBoost achieved the highest Area Under the Curve (AUC) value of 0.943, followed closely by Random Forest with an AUC of 0.942, while the neural network achieved an AUC of 0.940. The high AUC values across all models indicate strong discriminative ability between urban and non-urban classes.

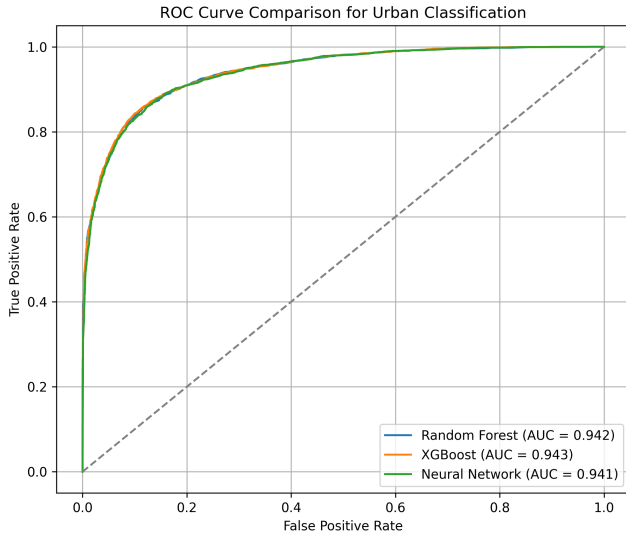


Figure 1: ROC curves comparing Random Forest, XGBoost, and neural network models, showing consistently high discriminative performance (AUC > 0.94), with XGBoost achieving the highest AUC.

The comparable performance across Random Forest, XGBoost, and the neural network suggests that feature representation plays a more critical role than model complexity in

multisource geospatial classification tasks. This indicates that when heterogeneous data sources are well integrated, simpler ensemble methods can achieve performance comparable to more complex deep learning models.

3.2 Feature Importance

The relative importance of predictor variables derived from the Random Forest model is presented in Figure 2. The results show that the Green spectral band contributes most to the classification, followed by night-time lights and population density, suggesting that both spectral reflectance and human activity indicators play key roles in distinguishing urban from non-urban areas.

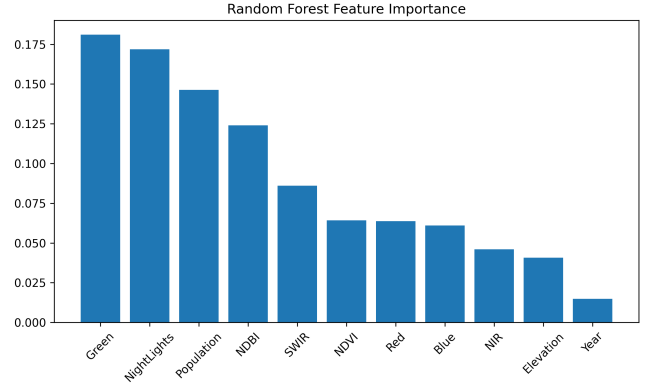


Figure 2: Random Forest feature importance showing that the Green spectral band, night-time lights, and population density are the most influential predictors of urban classification.

3.3 SHAP Explainable AI Results

To further interpret the model predictions, SHAP analysis was conducted. The SHAP summary (beeswarm) plot in Figure 3 illustrates the overall influence of each predictor variable on urban classification.

The results show that Night-time Lights produced the highest SHAP values, indicating that higher light intensity strongly increases the likelihood of a location being classified as urban. Population density also demonstrated a strong positive relationship with urban classification. Conversely, NDVI values exhibited negative SHAP contributions, suggesting that areas with higher vegetation density are more likely to be classified as non-urban.

The SHAP feature importance bar plot (Figure 4) confirms the dominant role of night-time lights and population density in the classification process. The dependence plot for night-time lights (Figure 5) further illustrates how increasing light intensity corresponds to higher SHAP values, reinforcing its strong association with urban development.

From an interpretability perspective, these results highlight the importance of combining predictive modeling with feature attribution methods to uncover domain-relevant patterns. The consistency between SHAP-derived feature importance and known urban indicators suggests that explainable AI can provide reliable insights into complex geospatial processes.

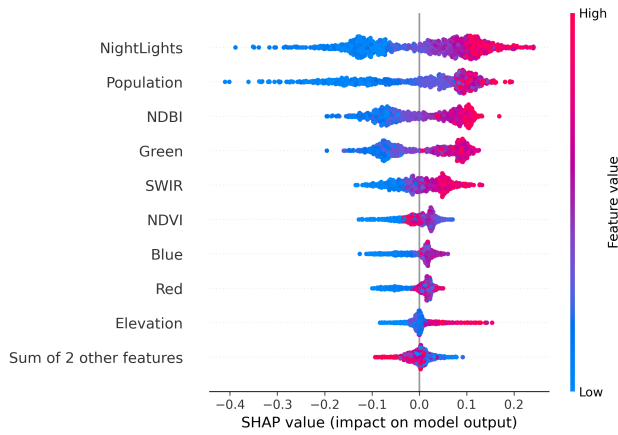


Figure 3: SHAP summary plot illustrating the global influence of predictor variables on urban classification, with night-time lights and population density showing strong positive contributions, while NDVI exhibits negative effects.

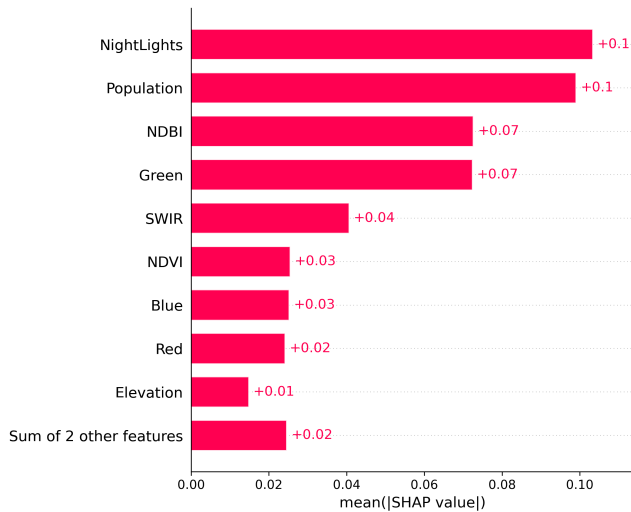


Figure 4: SHAP feature importance based on mean absolute SHAP values, confirming the dominant role of night-time lights and population density in urban classification.

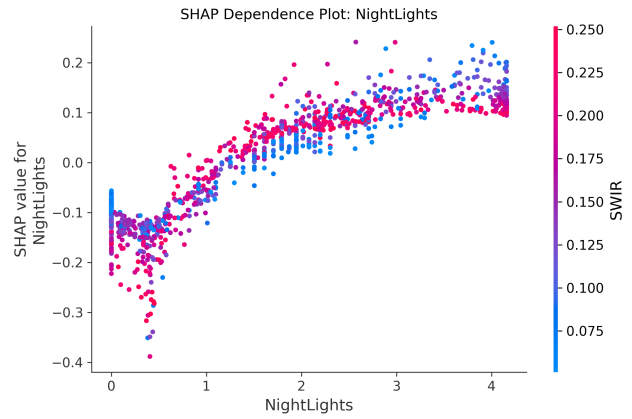


Figure 5: SHAP dependence plot for night-time lights showing a strong positive relationship between light intensity and the likelihood of urban classification.

3.4 Key Findings

Overall, the results demonstrate that all three models achieved consistently high performance in urban classification, with XGBoost showing a slight advantage in AUC. The strong discriminatory power of the multisource geospatial dataset highlights the complementary roles of spectral and socio-economic variables, with the Green spectral band emerging as the most influential predictor, followed by night-time lights and population density. This suggests that both surface reflectance characteristics and human activity indicators are critical for distinguishing urban areas. SHAP-based explainability further clarifies model behavior, revealing a negative association between vegetation (NDVI) and urban areas, and a positive contribution from built-up indices such as NDBI. Collectively, these findings emphasize the robustness and interpretability of the proposed approach, demonstrating its effectiveness for accurate and scalable urban mapping in rapidly growing cities.

3.5 Spatial Distribution of Urban Areas

The spatial distribution of predicted urban areas in Lagos for 2020 is shown in Figure 6. The map was generated using the XGBoost model, which achieved the highest classification performance among the evaluated models. The results reveal a clear concentration of urban areas in central Lagos, with expansion extending into surrounding peri-urban regions. The spatial pattern highlights the heterogeneous nature of urban development, with varying intensities of urbanization across the study area.

4 Discussion

The results demonstrate that machine learning and deep learning approaches can effectively classify urban and non-urban areas using multisource geospatial data. All three models: Random Forest, XGBoost, and the neural network, achieved similar performance, with an accuracy of 0.87 and AUC values above 0.94. These findings are consistent with previous studies showing that ensemble tree-based models

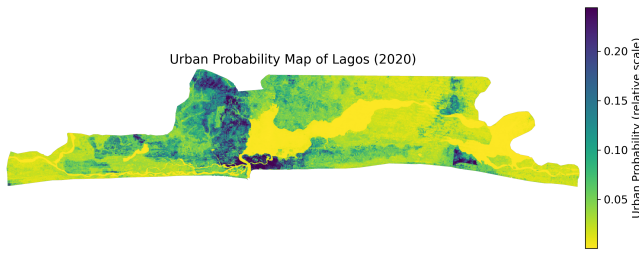


Figure 6: Urban probability map of Lagos (2020) generated using the proposed model. Higher values indicate a greater likelihood of urban presence. Probabilities are visualized using a relative scaling (2–98 percentile stretch) to enhance spatial contrast.

perform strongly in remote sensing classification tasks due to their ability to model complex nonlinear relationships [Belgiu and Drăguț, 2016].

The comparable performance between the machine learning models and the neural network suggests that the selected predictor variables provide strong discriminatory power for urban detection. While deep learning models often outperform traditional approaches when spatial context or image-based features are used, tree-based ensemble methods remain highly competitive when working with tabular datasets derived from spectral and socio-economic variables [Du K-L, 2025].

These findings emphasize that, in tabular geospatial settings, the quality and diversity of input features may outweigh the choice of model architecture. This aligns with broader observations in machine learning that data representation is often the primary driver of model performance, particularly when heterogeneous data sources are effectively combined.

The explainable AI analysis provided valuable insights into the drivers of urban classification. Night-time lights emerged as the most influential predictor, highlighting the strong relationship between artificial lighting intensity and urban development. This aligns with previous studies where night-time lights serve as proxies for urbanization and economic activity. Population density also played a significant role, reinforcing the importance of integrating socio-economic data with remote sensing variables to improve classification performance [F. Zhao and Xie, 2021].

More broadly, the integration of explainable AI within geospatial modeling frameworks enables a shift from purely predictive systems to decision-support tools that provide actionable insights. This is particularly important in urban studies, where transparency and interpretability are essential for policy and planning applications.

Spectral indices such as NDBI contributed substantially to the classification results by effectively capturing built-up surface characteristics, while NDVI exhibited a negative relationship with urban classification due to its association with vegetation. These findings confirm the complementary roles of spectral and socio-economic variables in urban detection.

The prominence of the Green spectral band as the most influential predictor further highlights the importance of surface reflectance characteristics in urban classification. Unlike derived indices such as NDVI and NDBI, the Green band re-

tains raw spectral information that captures subtle variations between built-up surfaces, vegetation, and transitional land cover types. In heterogeneous urban environments such as Lagos, where built-up areas are often interspersed with vegetation and bare soil, this band provides enhanced sensitivity to mixed land-use patterns. The high importance of the Green band therefore suggests that spectral reflectance differences play a critical role in distinguishing complex urban landscapes, complementing socio-economic indicators such as night-time lights and population density.

The SHAP analysis further demonstrated the value of explainable AI techniques in geospatial modeling by providing transparent insights into model behavior. This level of interpretability is particularly important for urban planning applications, where understanding the drivers of spatial patterns is as critical as achieving high predictive accuracy.

The spatial prediction map provides additional insight into the distribution of urban areas across Lagos. The results reveal a pronounced concentration of urban development within the metropolitan core, with expansion extending into surrounding peri-urban regions. The probability-based representation highlights spatial gradients of urban intensity, indicating that urbanization is heterogeneous rather than uniform across the landscape. These spatial patterns reflect typical outward expansion in rapidly growing cities and underscore the importance of integrating spatial outputs with model evaluation metrics.

Overall, while the performance metrics confirm strong classification capability, the spatial outputs reveal the geographic structure and heterogeneity of urban growth. This demonstrates the added value of combining machine learning, multisource data, and explainable AI to achieve both accurate and interpretable urban mapping.

A limitation of this study is that spatial context was not explicitly modeled, which could be explored in future work using convolutional neural networks or spatially explicit deep learning approaches.

5 Conclusion

This study presents an interpretable machine learning framework for urban classification using multisource geospatial data in Lagos, Nigeria. By integrating spectral, socio-economic, and temporal features, the proposed approach achieves strong predictive performance across multiple models, with accuracy of 0.87 and AUC values exceeding 0.94. The comparable performance of ensemble methods and the neural network highlights the effectiveness of tabular multisource data for urban detection.

Beyond predictive accuracy, the incorporation of SHAP-based explainable AI provides clear insights into the drivers of urban classification. Night-time lights and population density emerge as dominant predictors, while spectral variables contribute complementary information for distinguishing complex urban environments. These results demonstrate that combining heterogeneous data sources with interpretable machine learning enables both accurate and transparent urban mapping.

The proposed framework offers a scalable and practical

tool for monitoring urban growth in rapidly developing regions. Future work will explore the integration of spatial context through convolutional and spatially explicit deep learning models, as well as the extension of the framework to other cities and data environments. More broadly, this work demonstrates how integrating interpretable machine learning with heterogeneous data sources can serve as a general paradigm for geospatial AI applications.

References

- [A. V. Ponce-Bobadilla and Stodtmann, 2024] C. S. Maier S. Mensing A. V. Ponce-Bobadilla, V. Schmitt and S. Stodtmann. Practical guide to shap analysis: Explaining supervised machine learning model predictions in drug development. *Clinical and Translational Science*, 2024.
- [Belgiu and Drăguț, 2016] Mariana Belgiu and Lucian Drăguț. Random forest in remote sensing: A review of applications and future directions. *ISPRS Journal of Photogrammetry and Remote Sensing*, 114:24–31, 2016.
- [Chen and Guestrin, 2016] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, New York, NY, USA, 2016. Association for Computing Machinery.
- [Du K-L, 2025] Jiang B Zeng J Lu J Du K-L, Zhang R. Foundations and innovations in data fusion and ensemble learning for effective consensus. *Mathematics*, 13(4):587, 2025.
- [F. Zhao and Xie, 2021] Z. Peng J. Yang G. Luan C. Chu J. Ding S. Feng Y. Jing F. Zhao, L. Song and Z. Xie. Night-time light remote sensing mapping: Construction and analysis of ethnic minority development index. *Remote Sensing*, 13(11):2129, 2021.
- [Fawcett, 2005] Tom Fawcett. An introduction to roc analysis. *Pattern Recognition Letters*, 27(8):861–874, 2005.
- [L. Ma and Johnson, 2019] X. Zhang Y. Ye G. Yin L. Ma, Y. Liu and B. A. Johnson. Deep learning in remote sensing applications: A meta-analysis and review. *ISPRS Journal of Photogrammetry and Remote Sensing*, 152:166–177, 2019.
- [Li and Zhou, 2016] X. Li and Y. Zhou. Urban mapping using dmsp/ols stable night-time light: A review. *International Journal of Remote Sensing*, 38(21):6030–6042, 2016.
- [Lundberg and Lee., 2017] Scott Lundberg and Su-In Lee. Unified approach to interpreting model predictions. *arXiv preprint arXiv:1705.07874*, 2017.
- [Seto *et al.*, 2012] Karen C. Seto, Burak Güneralp, and Lucy R. Hutya. Global forecasts of urban expansion to 2030 and direct impacts on biodiversity and carbon pools. *Proceedings of the National Academy of Sciences*, 109(40):16083–16088, 2012.
- [T. Man and Ignatov, 2024] N. Zhukova A. Subbotin T. Man, V. Yu. Osipov and D. I. Ignatov. Neural networks for intelligent multilevel control of artificial and natural objects based on data fusion: A survey. *Information Fusion*, 2024.
- [Tatem, 2017] Andrew J. Tatem. Worldpop, open data for spatial demography. *Scientific Data*, 4:170004, 2017.
- [United Nations Human Settlements Programme, 2022] United Nations Human Settlements Programme. World cities report 2022: Envisaging the future of cities. Technical report, UN-Habitat, 2022.
- [Weng, 2012] Qihao Weng. Remote sensing of impervious surfaces in the urban areas: Requirements, methods, and trends. *Remote Sensing of Environment*, 117:34–49, 2012.
- [Y. Zha and Ni, 2003] J. Gao Y. Zha and S. Ni. Use of normalized difference built-up index in automatically mapping urban areas from tm imagery. *International Journal of Remote Sensing*, 24(3):583–594, 2003.
- [Yu and Fang, 2023] D. Yu and C. Fang. A review and renewed perspective of urban studies in recent decades. *Remote Sensing*, 15(5):1307, 2023.
- [Z. Sun and Liang., 2024] P. Li H. Wang M. Zhang Z. Sun, G. Wang and X. Liang. An improved random forest based on the classification accuracy and correlation measurement of decision trees. *Expert Systems with Applications*, 2024.